

WARMUP

Calculate the slope between the two points :
(2,5) and (6, 26)



FUNDAMENTAL(BASE) UNITS IN PHYSICS

M- meters, m (length, position, distance, displacement)

K- kilogram, kg (mass)

S- seconds, (time)

Derived units – a combination of base units

Example: m/s, m/s/s

You will need to be able to convert units well.

KINEMATICS

Kinematics is the branch of mechanics that describes the motion of objects without necessarily discussing what causes the motion.

1-Dimensional Kinematics (or 1-Dimensional motion) refers to motion in a straight line.



THERE ARE TWO KINDS OF QUANTITIES...

Scalars are quantities that have magnitude only, such as

- distance
- Speed, average speed
- time
- mass

Vectors are quantities that have both magnitude and direction, such as

- position
- displacement
- Velocity, average velocity
- acceleration

NOTATING VECTORS

This is how you
notate a
vector...

$\mathbf{R}$

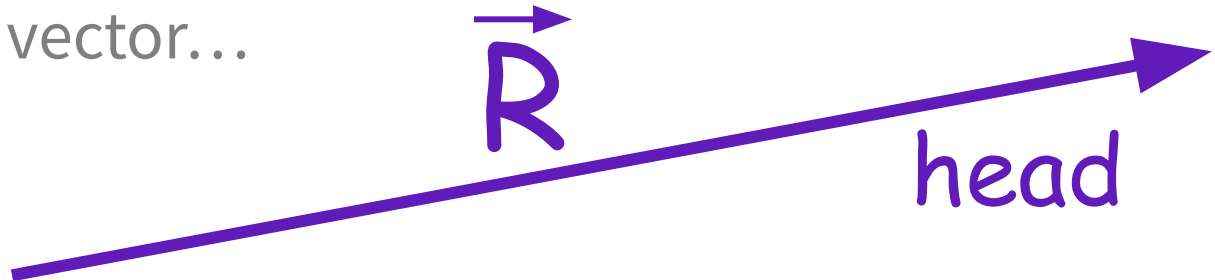
$\vec{\mathbf{R}}$

This is how you
draw a vector...

tail

$\vec{\mathbf{R}}$

head



DISTANCE

The total length of the path traveled by an object is called distance.

“How far have you walked?” is a typical distance question.

The SI (International system of units or metric system) unit of distance is the meter (m).



POSITION

The location of an object. (Also in meters, m)

DISPLACEMENT (Δx)

The change in the position of a particle is called displacement.

Δ is a Greek letter used to represent the words “change in”. Δx therefore means “change in x” or it could also mean “change in y”. It is always calculated by final value minus initial value.

“How far are you from home?” is a typical displacement question.

The displacement of an object can also be defined as the straight line distance between the starting and ending points with a particular direction.

The SI unit for displacement is the meter, m.

Calculation of displacement:

$$\Delta x = x_f - x_i = x - x_0$$

WARMUP

1. Convert 8.282 km to m
2. Convert 0.695 km/s to m/s
3. Identify the following as a scalar or vector
 - a. Mass c. acceleration
 - b. Displacement d. time
4. What is the speed of a car traveling at 70 m/s at 60 degrees N of W?

DISTANCE VS DISPLACEMENT

A person walks 75m east and an additional 40 east.
What is the person's distance? displacement?

DISTANCE VS DISPLACEMENT

A person walks 75 m east then 40 m west. What is the person's distance? displacement?

DISTANCE VS DISPLACEMENT

A person walks 75 m west and 40 m north. What is the person's distance? displacement?

AVERAGE SPEED

Average speed describes how fast a particle is moving. The equation is:

$$\bar{s} = \frac{d}{\Delta t}$$

Average speed is always a positive number.

- Where
 - $\bar{s}$ = average speed
 - d = distance
 - Δt = elapsed time
- The SI unit of speed is m/s

AVERAGE VELOCITY

Average velocity describes how fast the displacement is changing. The equation is:

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

Average velocity is + or – depending on direction.

- where:

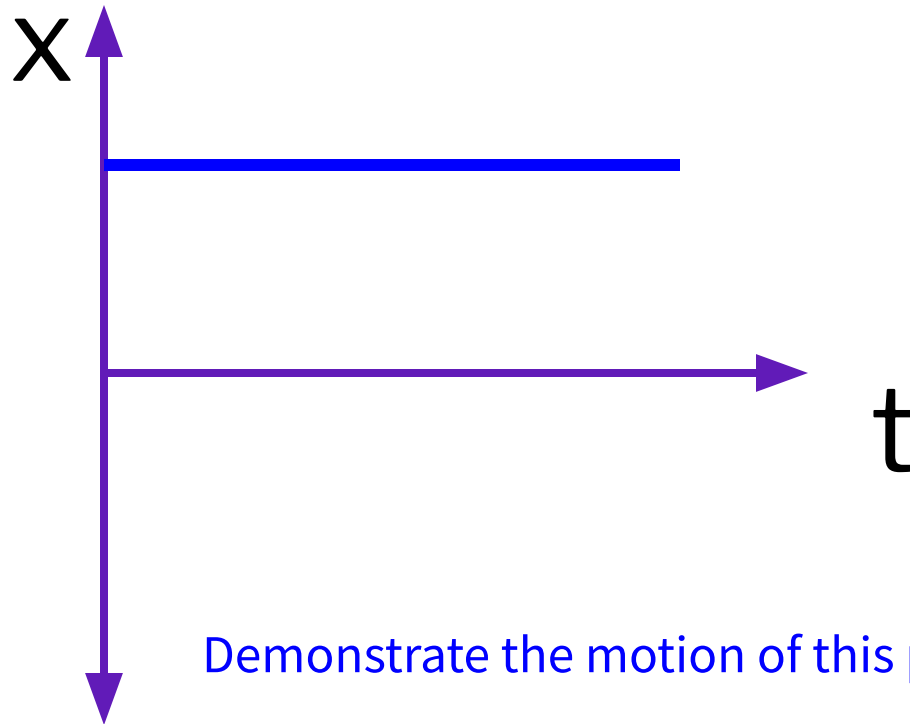
$\bar{v}$ = average velocity

Δx = displacement

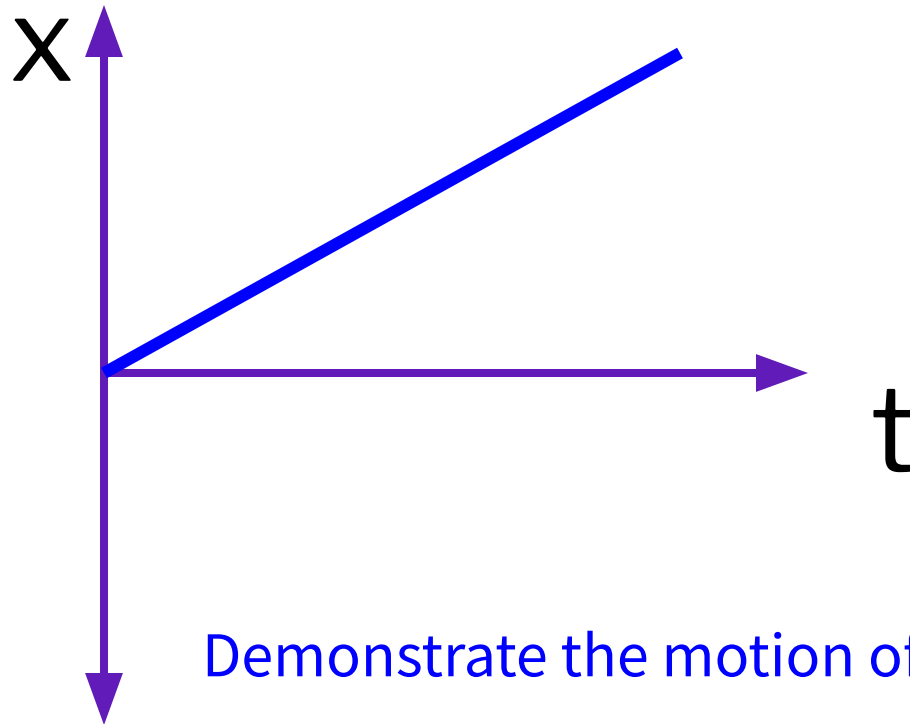
Δt = elapsed time

- The SI unit of velocity is m/s.

GRAPHICAL PROBLEM

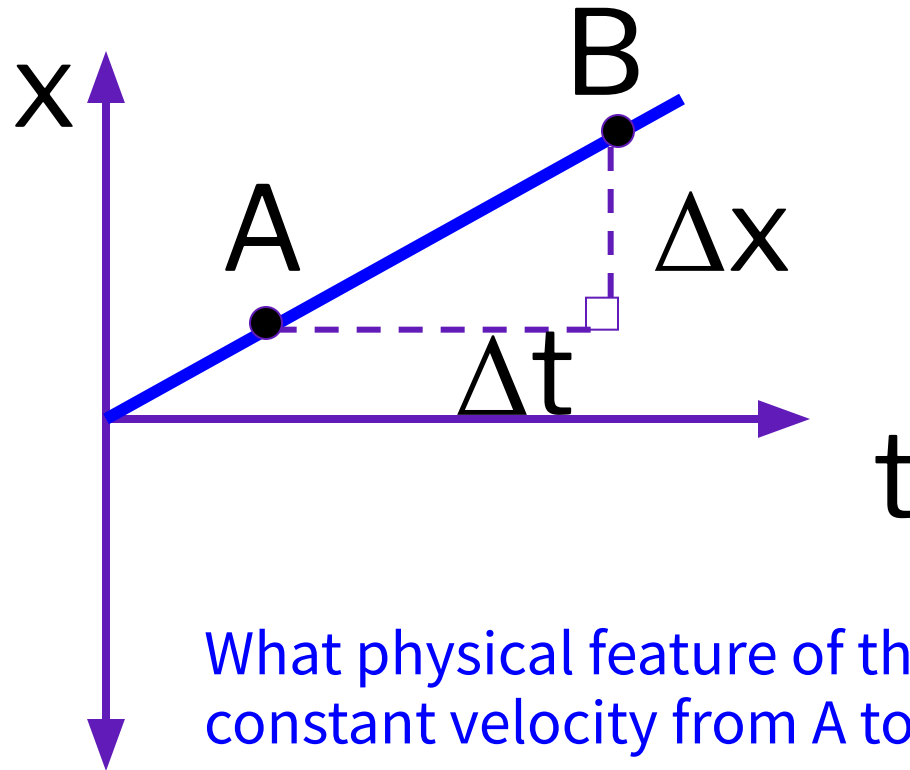


GRAPHICAL PROBLEM



Demonstrate the motion of this particle.

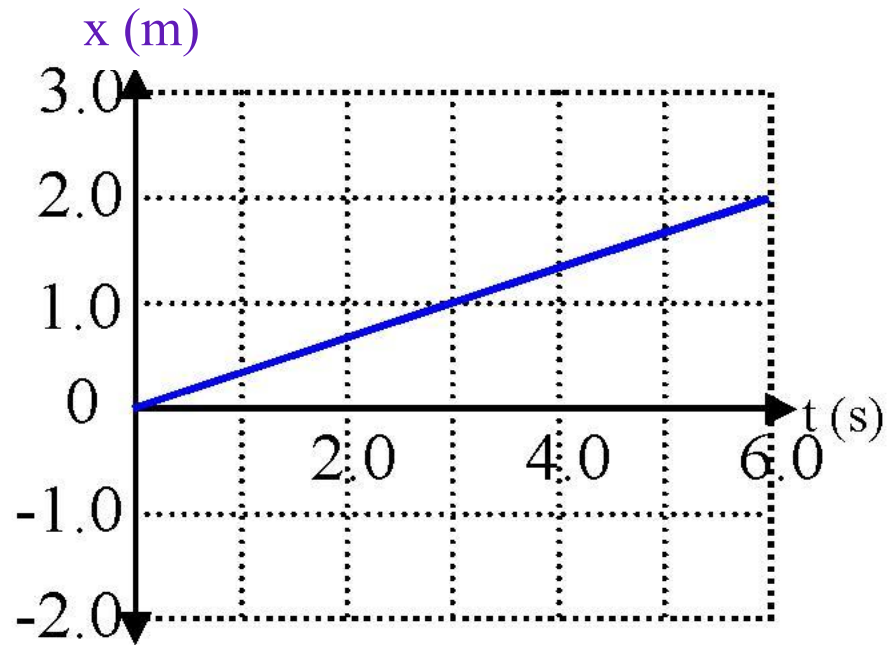
GRAPHICAL PROBLEM



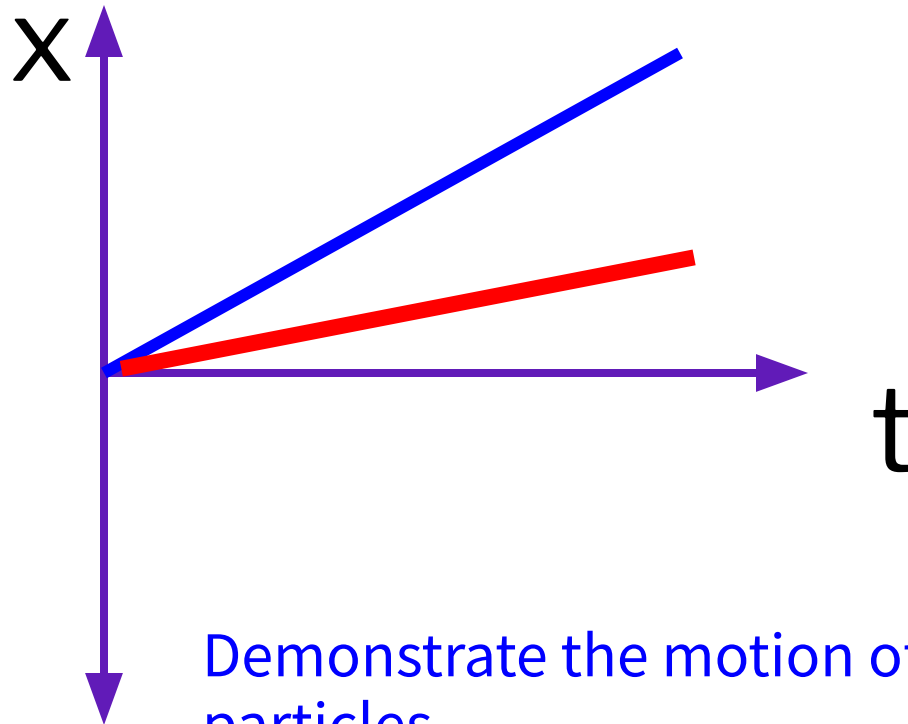
slope
 $v_{\text{ave}} = \Delta x / \Delta t$

What physical feature of the graph gives the constant velocity from A to B?

Graphical Problem: Determine the average velocity from the graph.

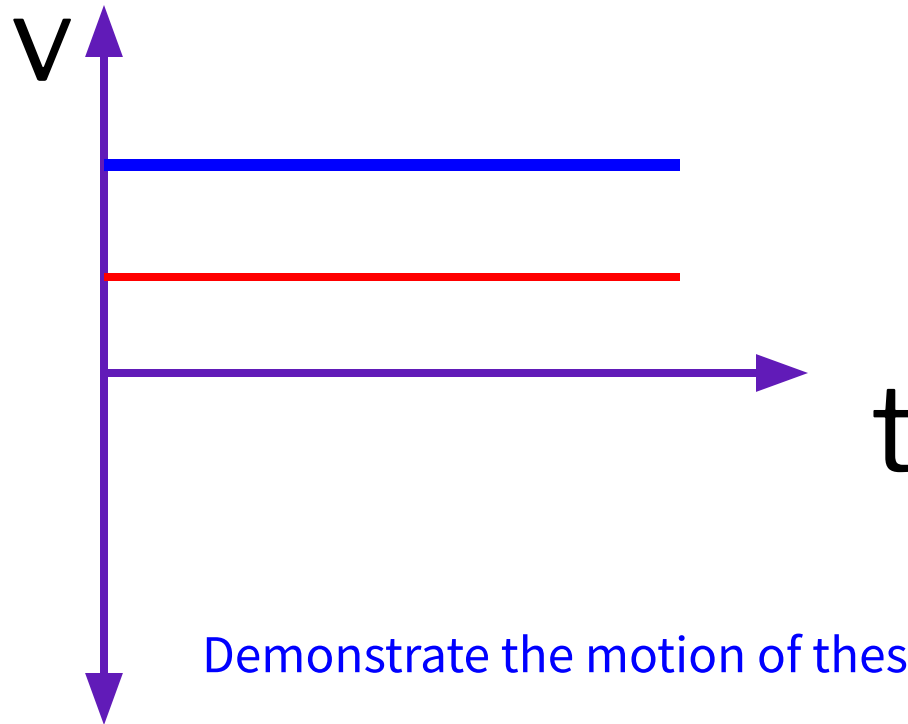


GRAPHICAL REVIEW PROBLEM



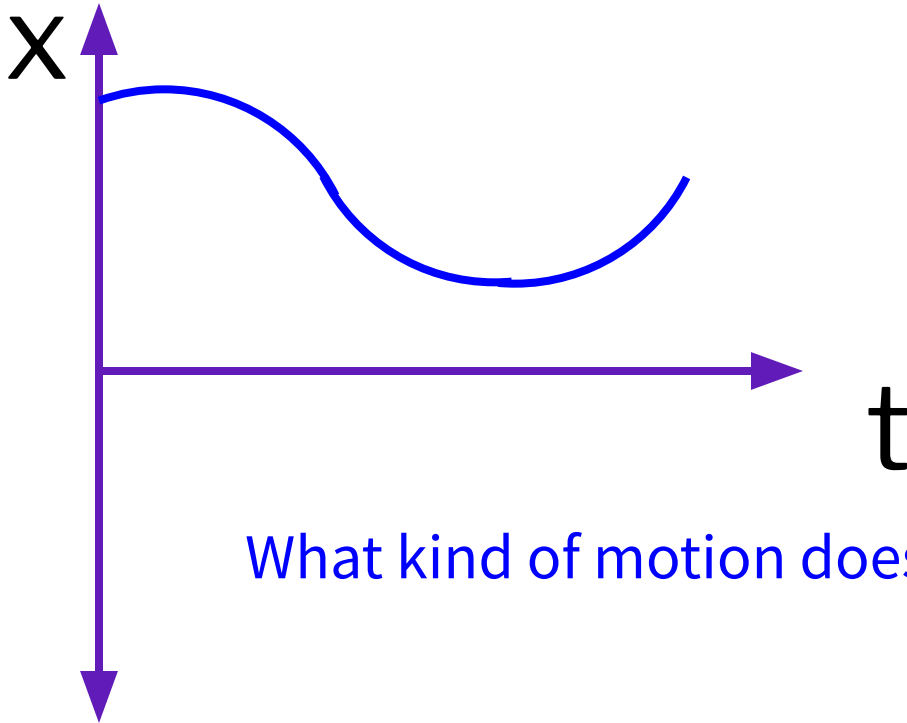
Demonstrate the motion of these two particles.

GRAPHICAL PROBLEM



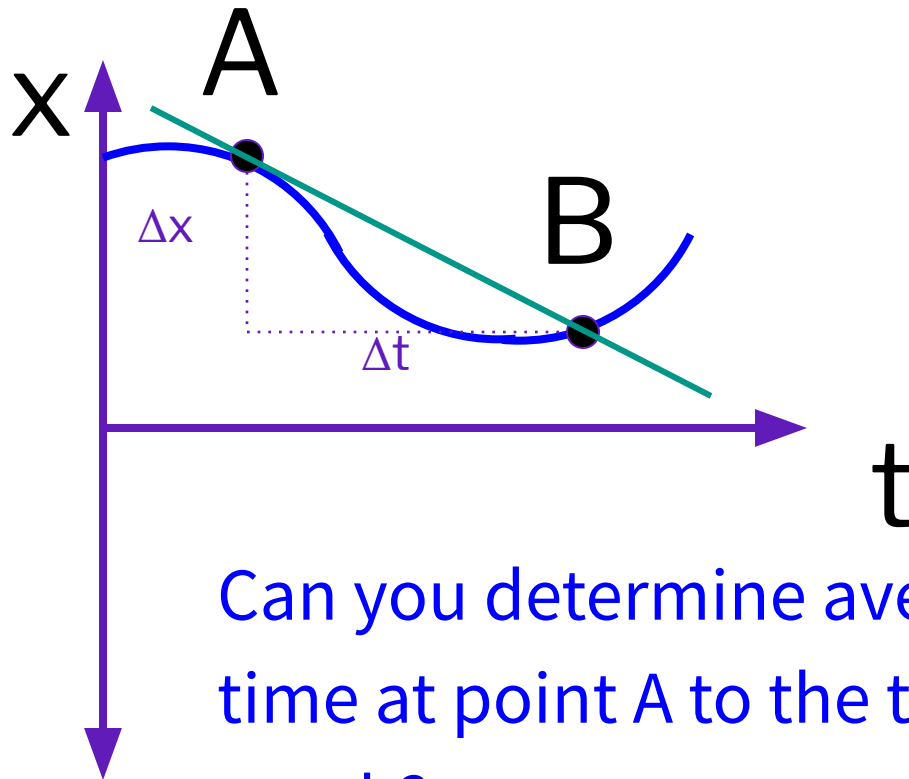
Demonstrate the motion of these two particle.

GRAPHICAL PROBLEM



What kind of motion does this graph represent?

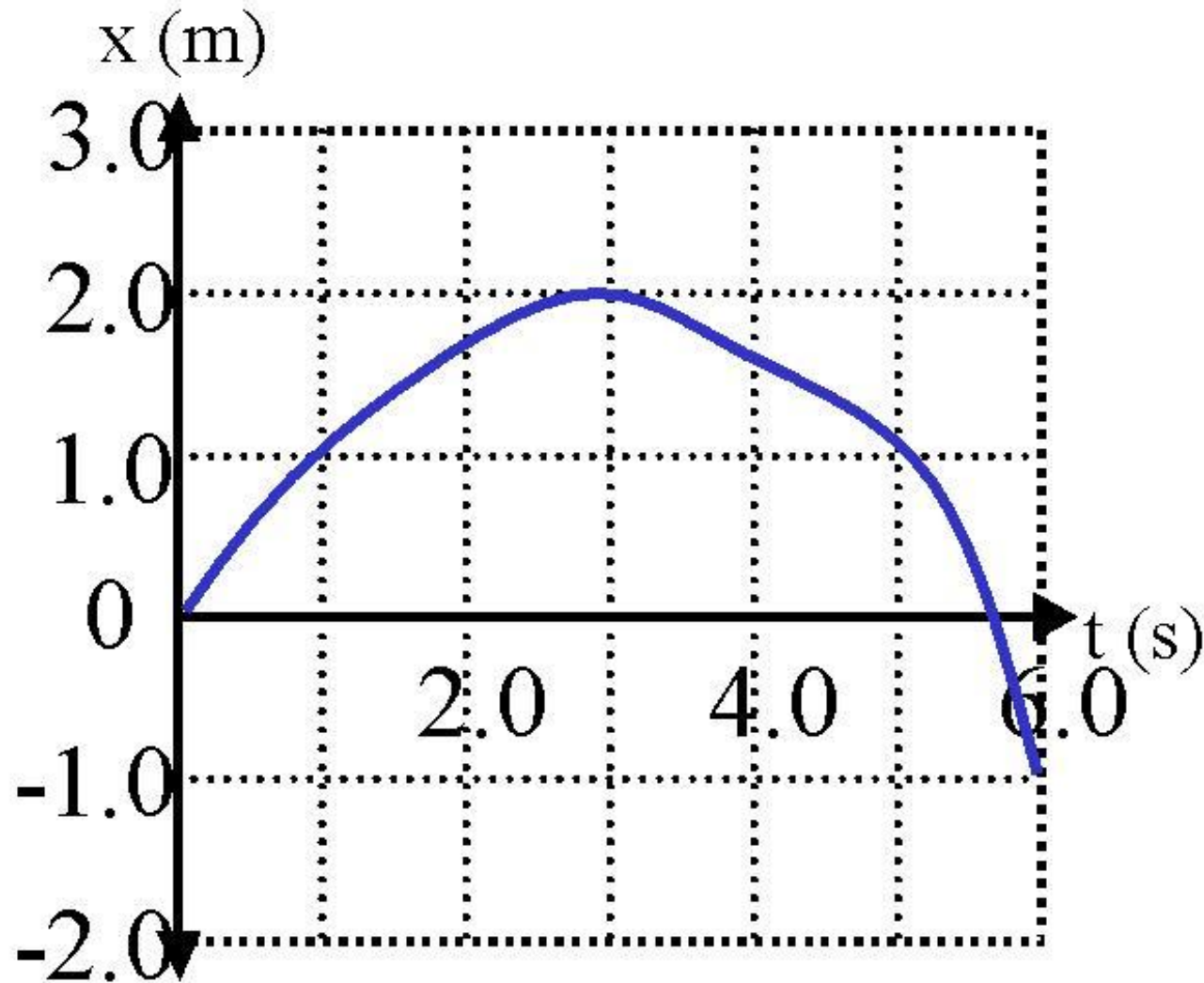
GRAPHICAL PROBLEM



$$v_{\text{ave}} = \Delta x / \Delta t$$

Can you determine average velocity from the time at point A to the time at point B from this graph?

Graphical Problem: Determine the average velocity between 1 and 6 seconds.



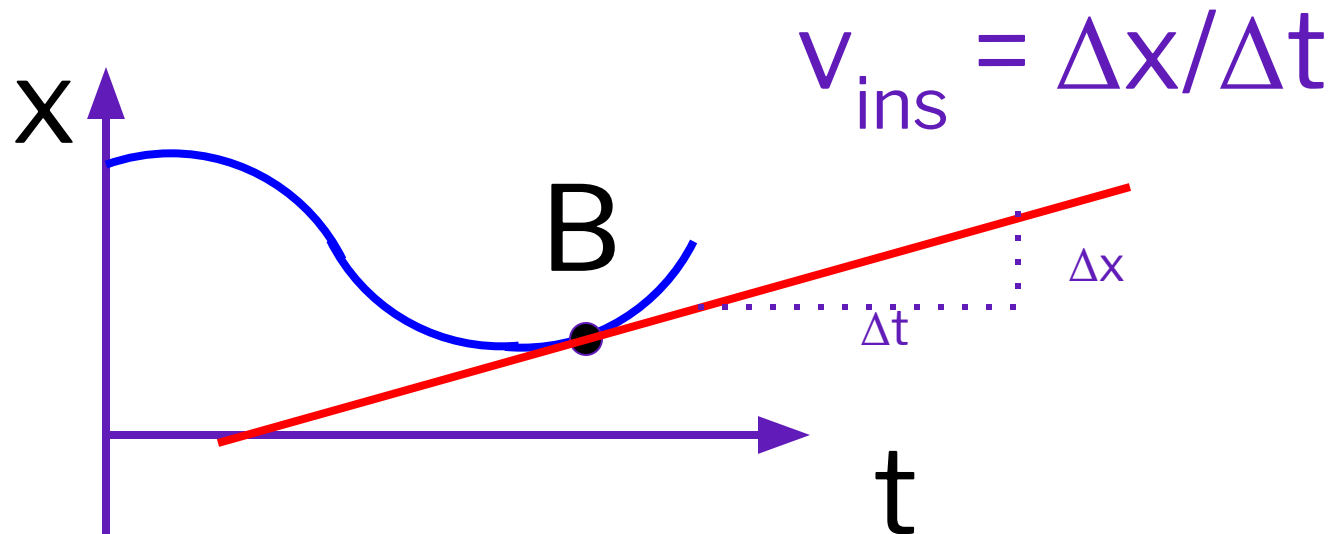
INSTANTANEOUS VELOCITY

The velocity at a single instant in time.

If the velocity is uniform, or constant, the instantaneous velocity is the same as the average velocity.

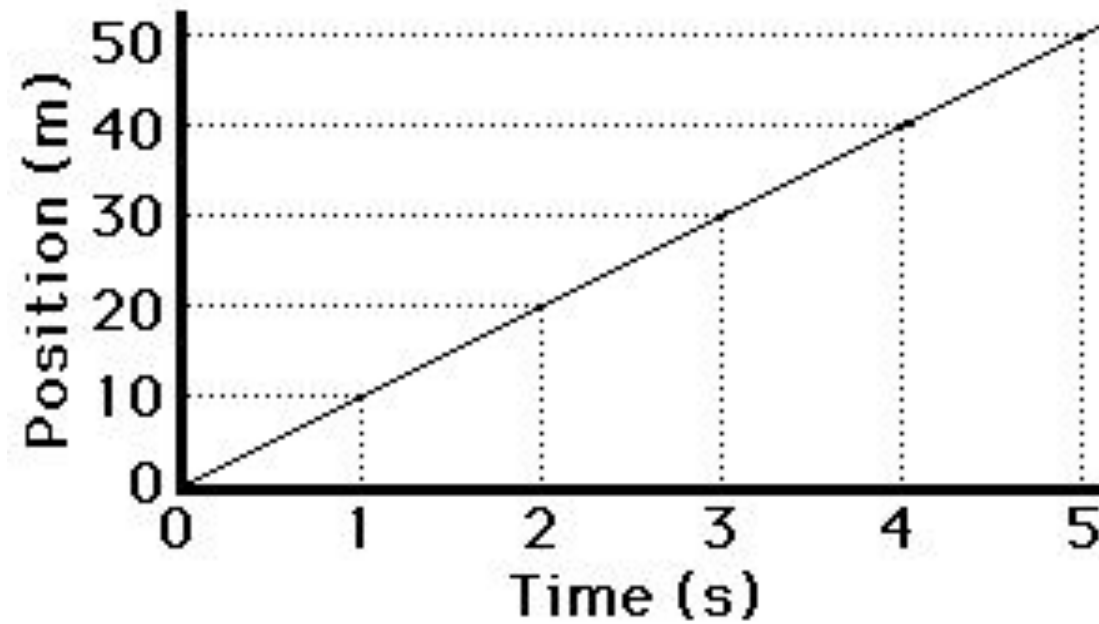
If the velocity is not constant, then the instantaneous velocity is not the same as the average velocity, and we must carefully distinguish between the two.

INSTANTANEOUS VELOCITY



Draw a tangent line to the curve at B. The slope of this line gives the instantaneous velocity at that specific time.

Practice Problem: Determine the instantaneous velocity at 1.0 second.



ACCELERATION (A)

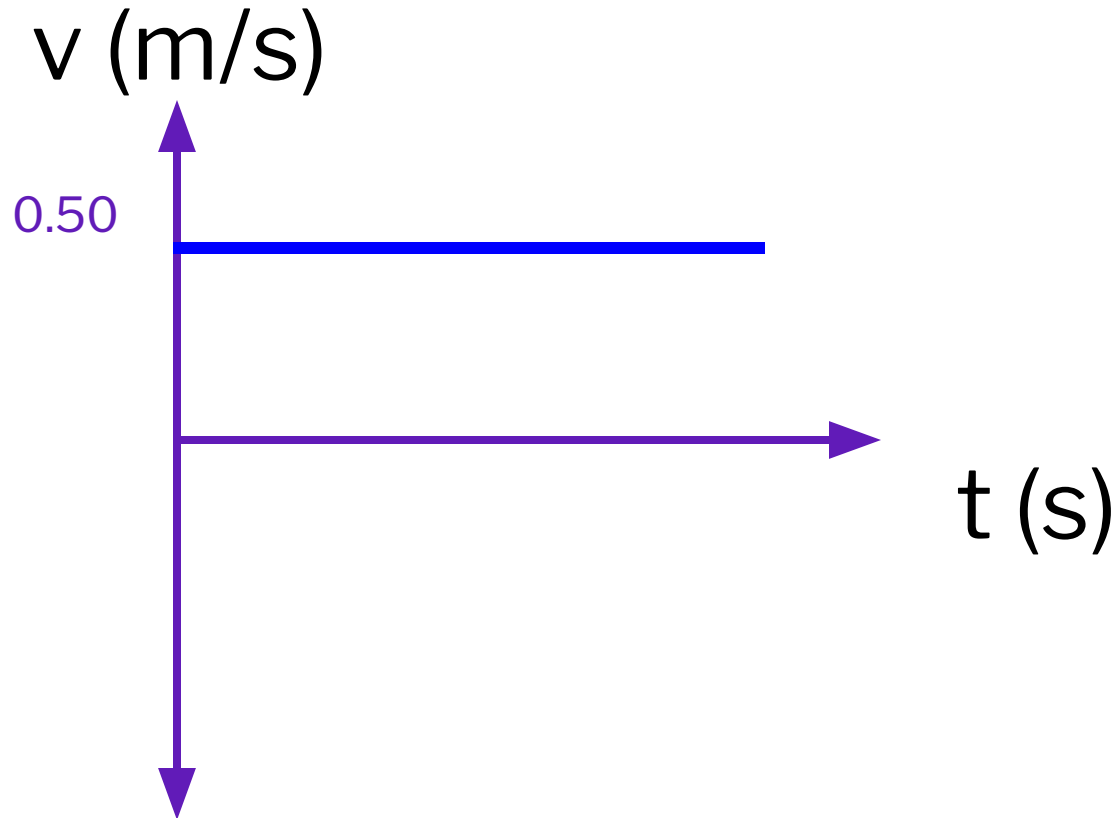
Any change in velocity over a period of time is called acceleration.

The sign (+ or -) of acceleration indicates its direction.

Acceleration can be...

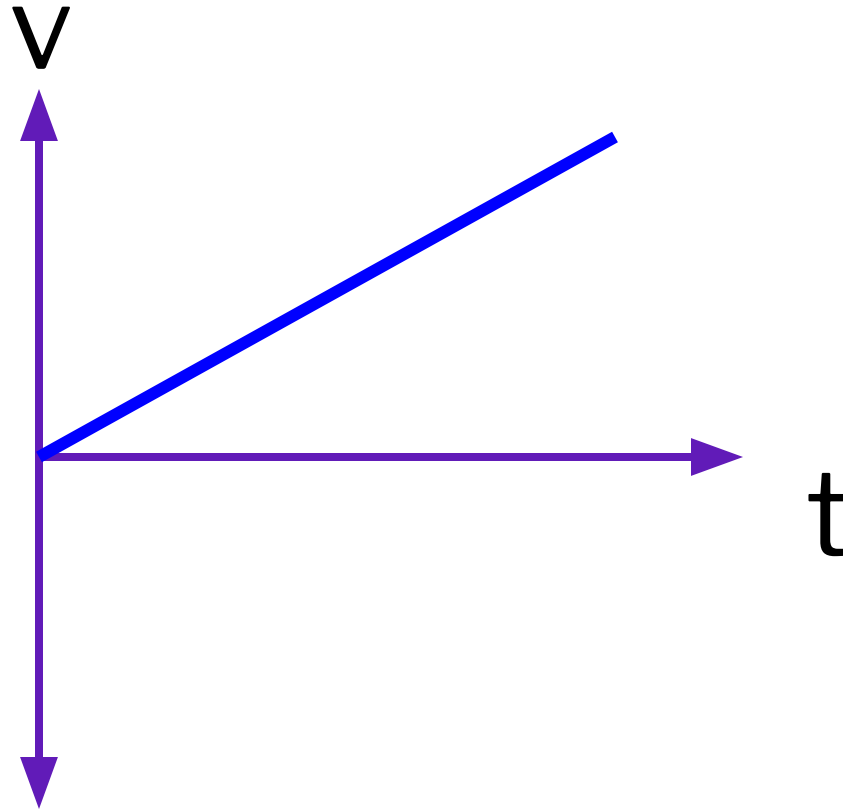
- speeding up
- slowing down
- turning

GRAPHICAL PROBLEM



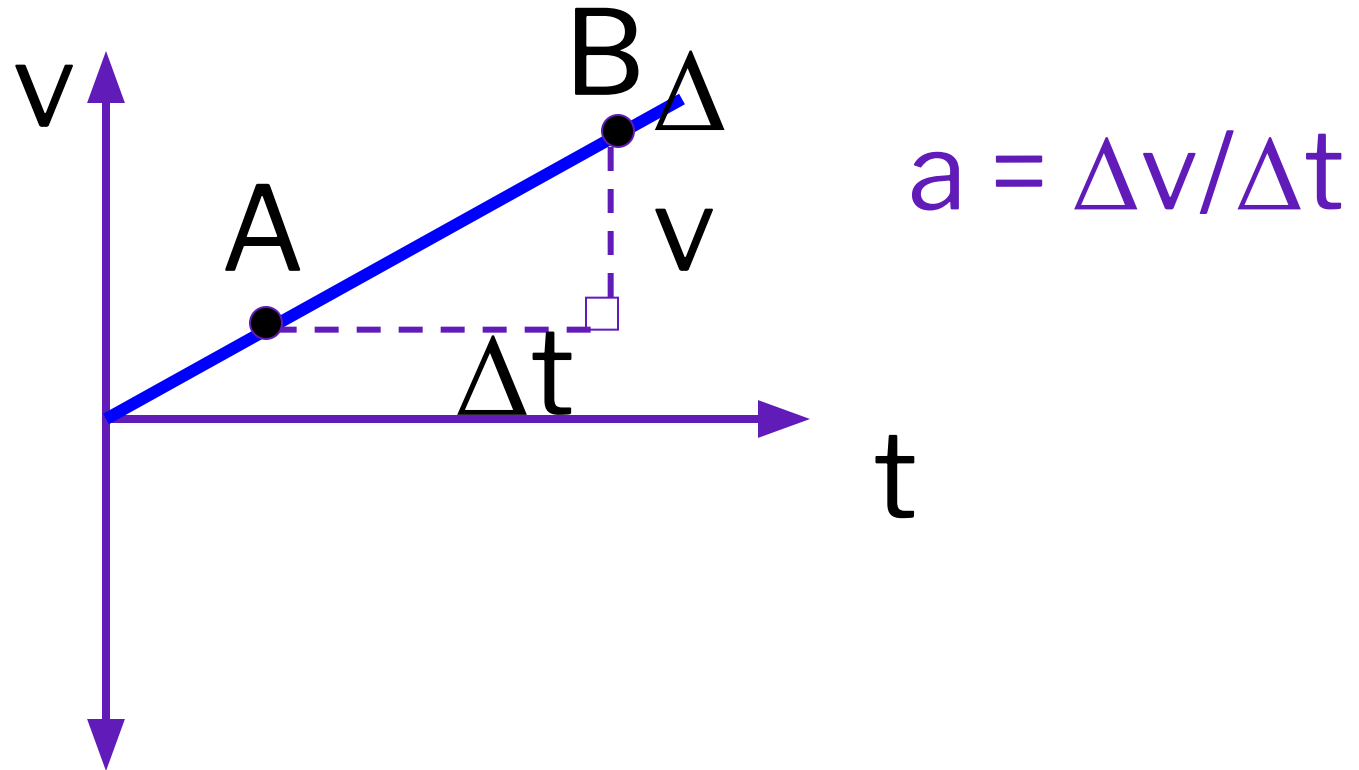
Demonstrate the motion of this particle. Is it accelerating?

GRAPHICAL PROBLEM



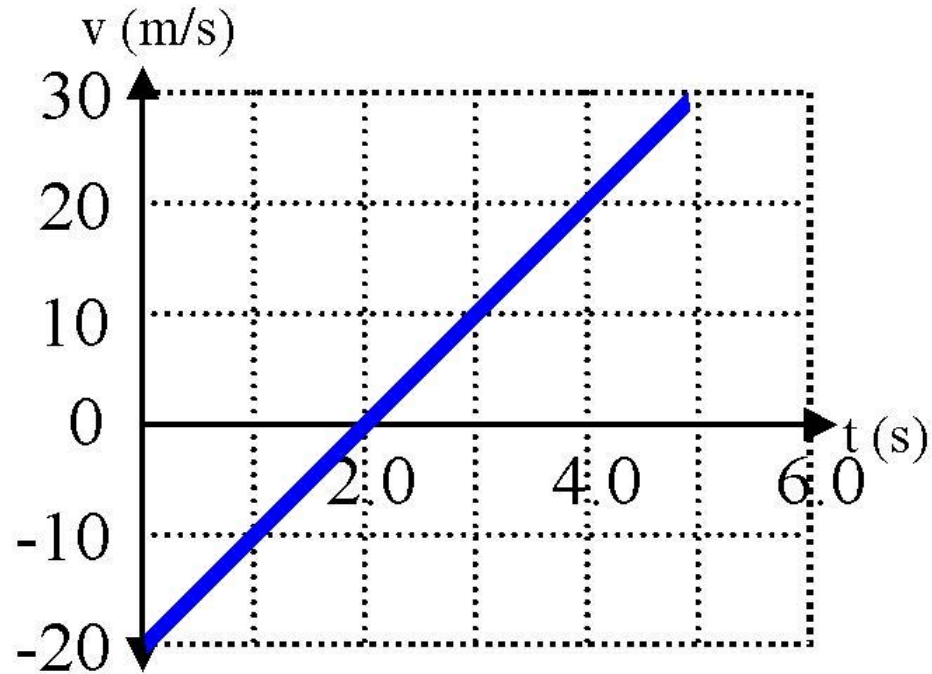
Demonstrate the motion of this particle. Is it accelerating?

GRAPHICAL PROBLEM

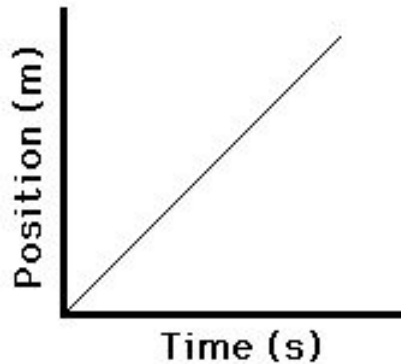


What physical feature of the graph gives the acceleration?

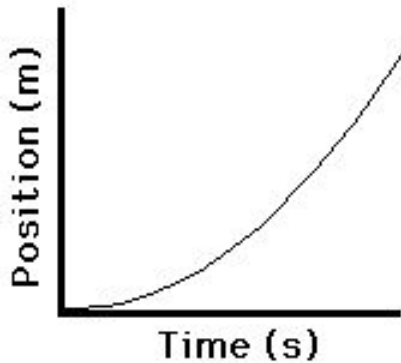
Practice Problem: Determine the acceleration from the graph.



POSITION VS TIME GRAPHS

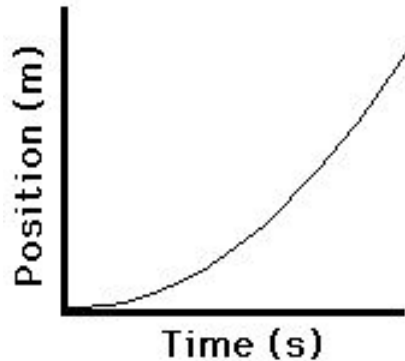


Particles moving with no acceleration (constant velocity) have graphs of position vs time with one slope. The velocity is not changing since the slope is constant.

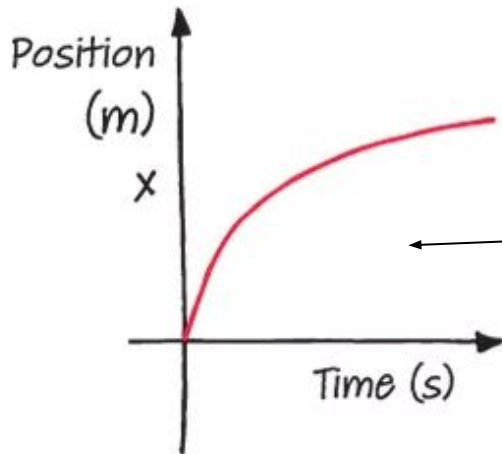


Position vs time graphs for particles moving with constant acceleration look parabolic. The slope is changing. In this graph it is increasing, and the particle is speeding up.

POSITION VS TIME GRAPHS-CONCAVITY

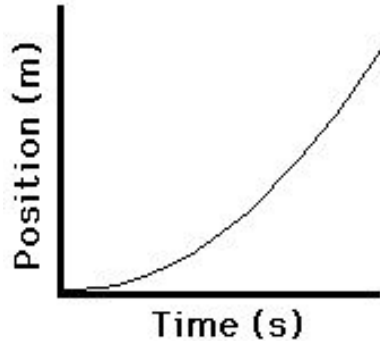


Position time graphs that open up show positive acceleration.

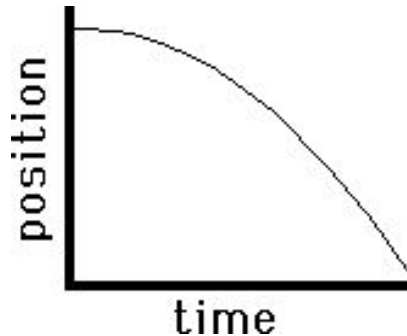


Position time graphs that open down show negative acceleration.

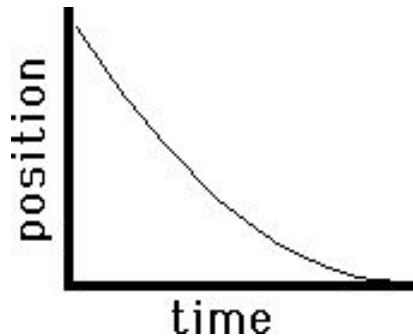
DESCRIBE THE MOTION



This object is moving in the positive direction and accelerating in the positive direction (speeding up).

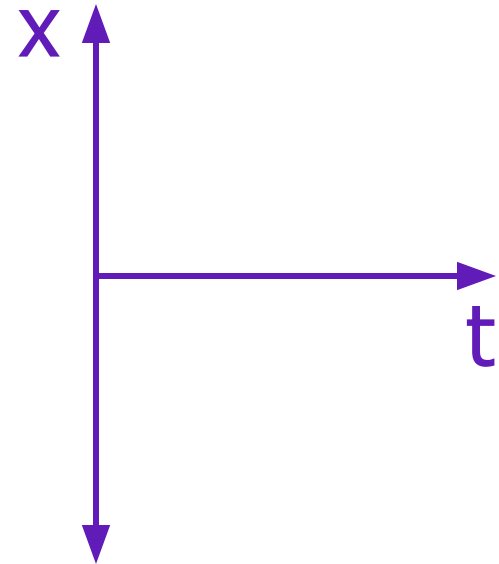


This object is moving in the negative direction and accelerating in the negative direction (speeding up).

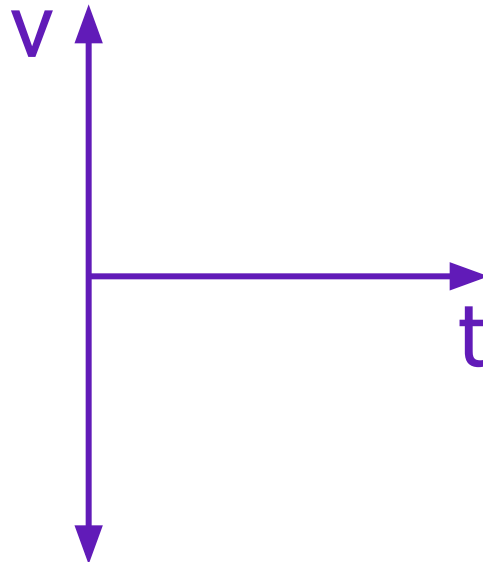


This object is moving in the negative direction and accelerating in the positive direction (slowing down).

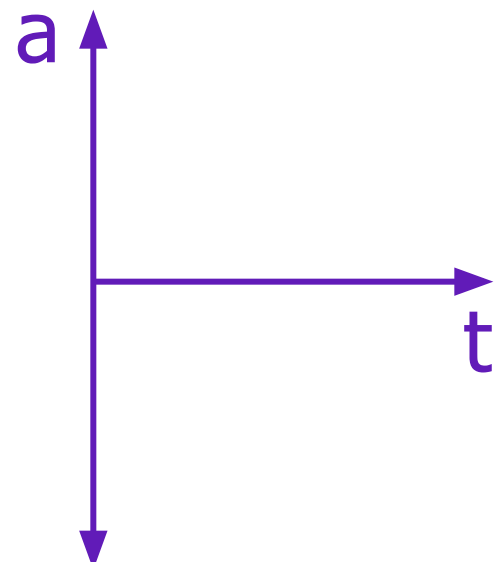
DRAW GRAPHS FOR STATIONARY PARTICLES



Position
VS
time

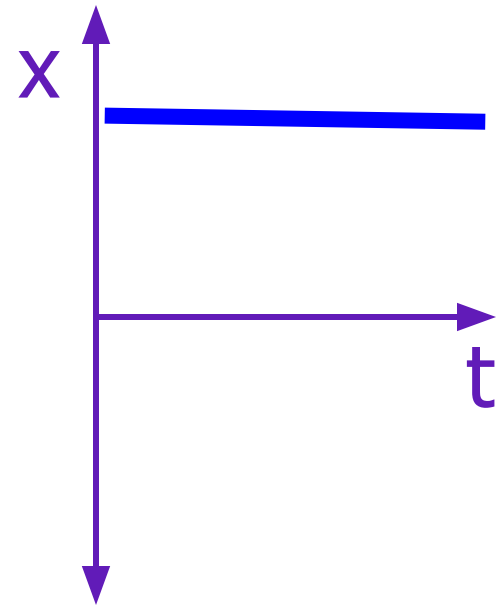


Velocity
VS
time

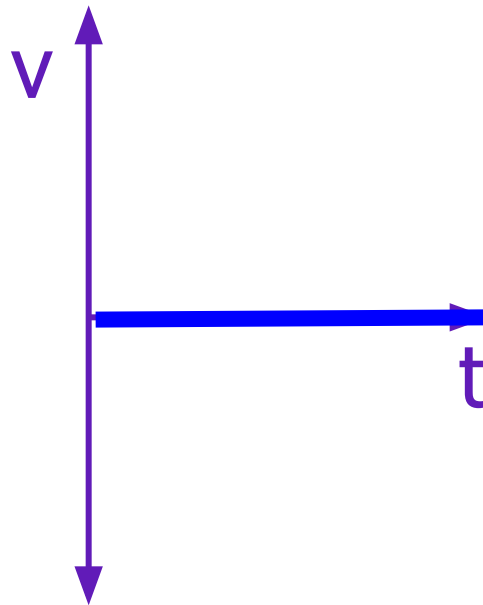


Acceleration
VS
time

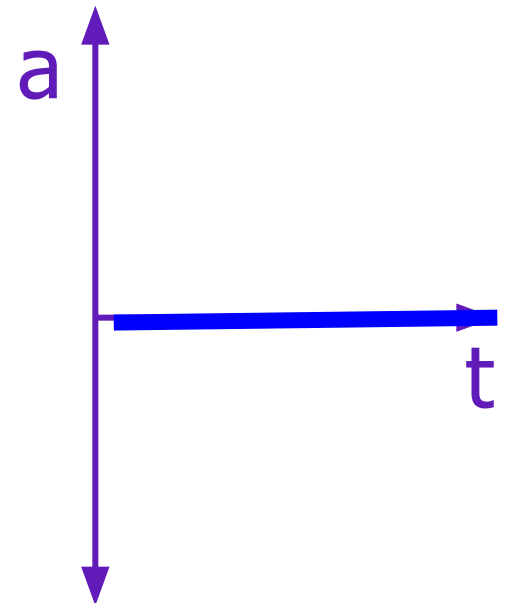
DRAW GRAPHS FOR STATIONARY PARTICLES



Position
vs
time

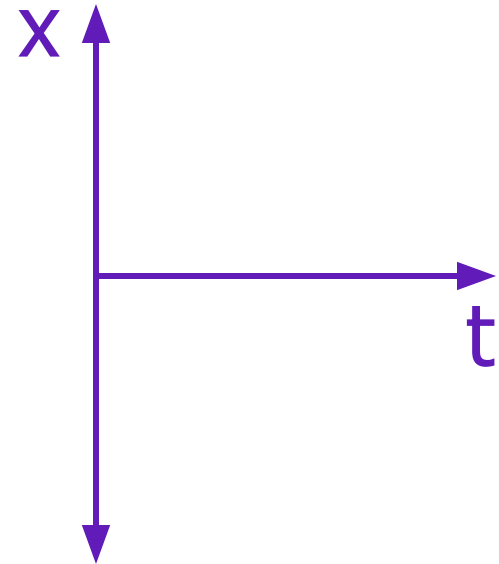


Velocity
vs
time

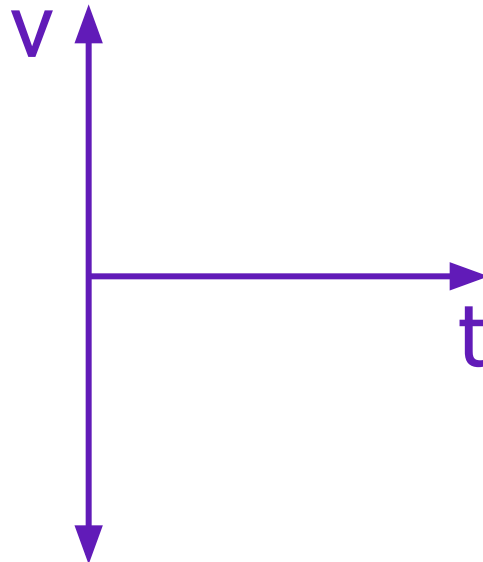


Acceleration
vs
time

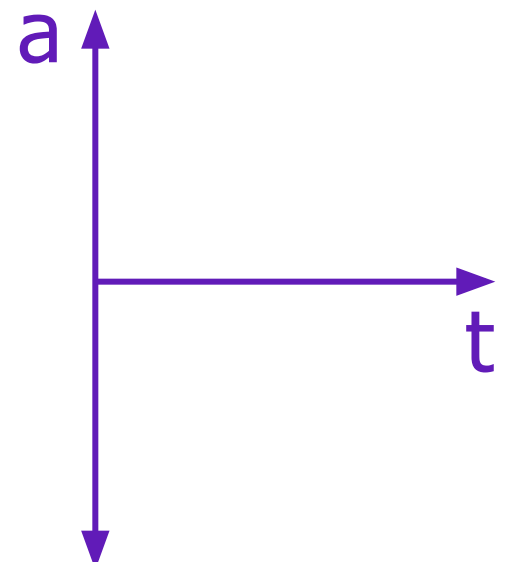
DRAW GRAPHS FOR CONSTANT NON-ZERO VELOCITY



Position
VS
time

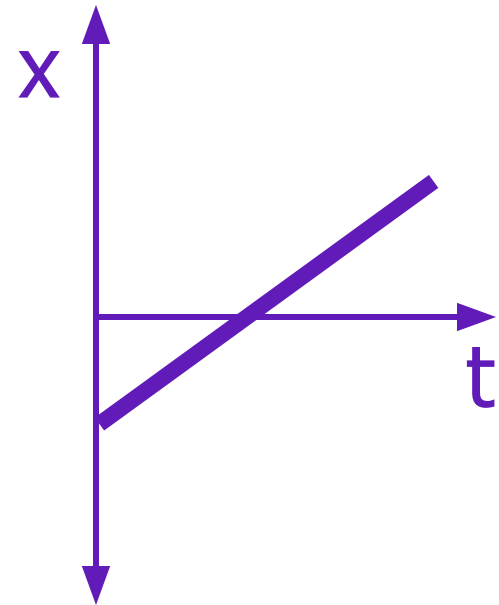


Velocity
VS
time

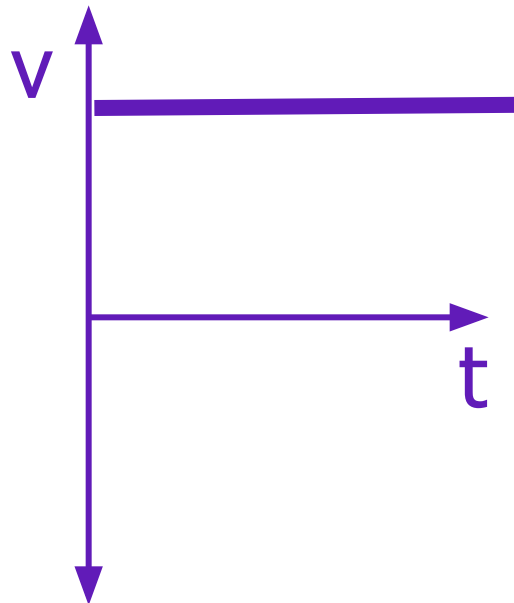


Acceleration
VS
time

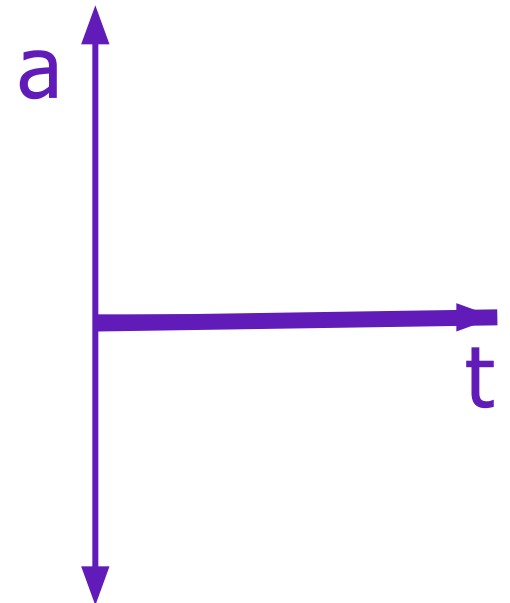
DRAW GRAPHS FOR CONSTANT NON-ZERO VELOCITY



Position
vs
time

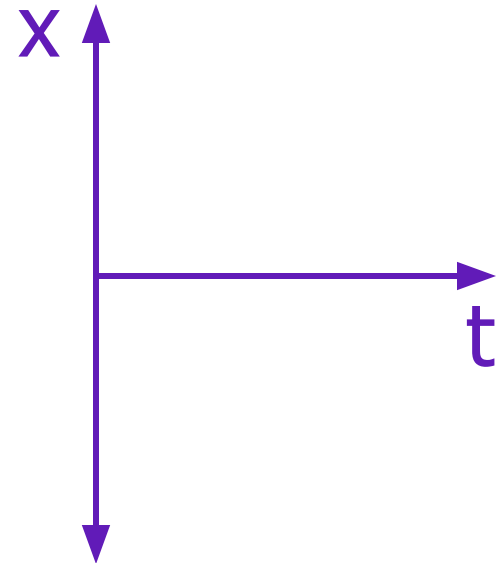


Velocity
vs
time

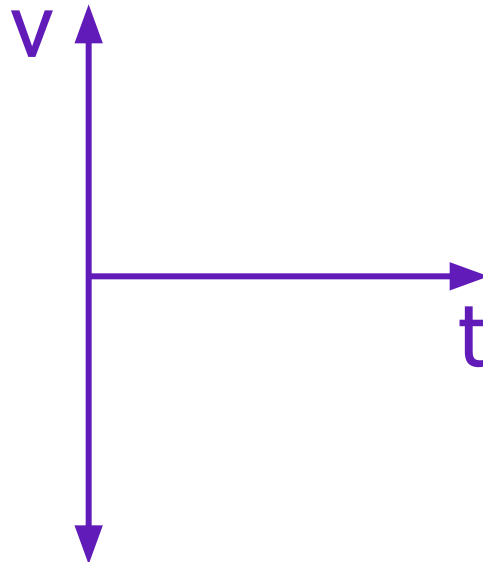


Acceleration
vs
time

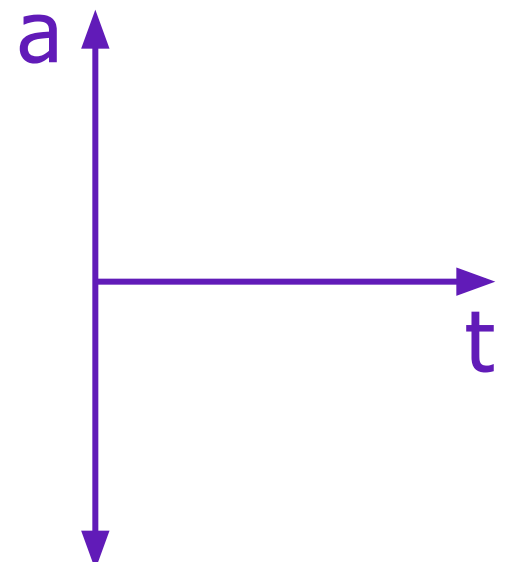
DRAW GRAPHS FOR CONSTANT NON-ZERO ACCELERATION



Position
VS
time

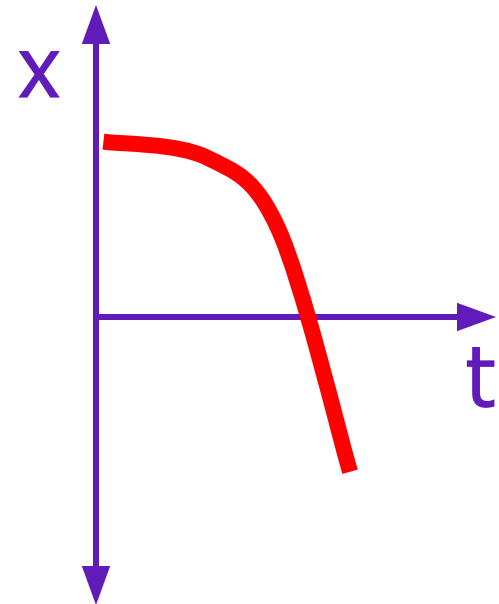


Velocity
VS
time

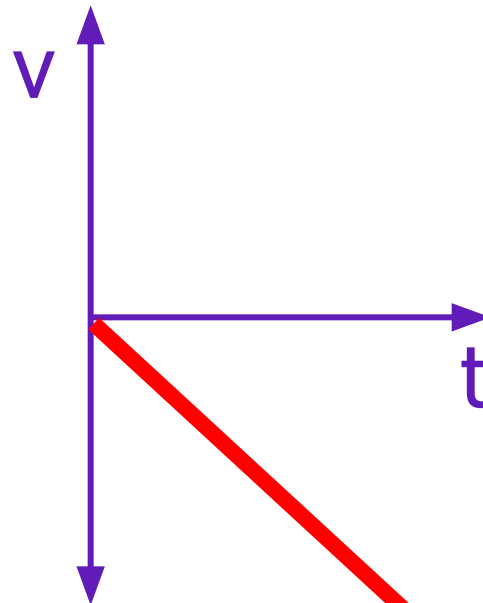


Acceleration
VS
time

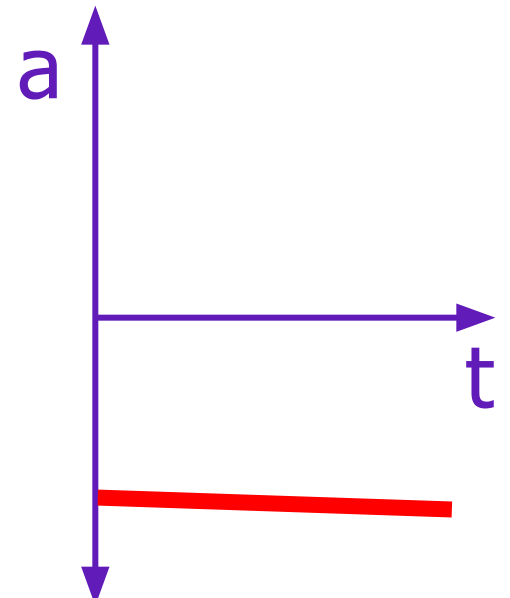
DRAW GRAPHS FOR CONSTANT NON-ZERO ACCELERATION



Position
vs
time



Velocity
vs
time



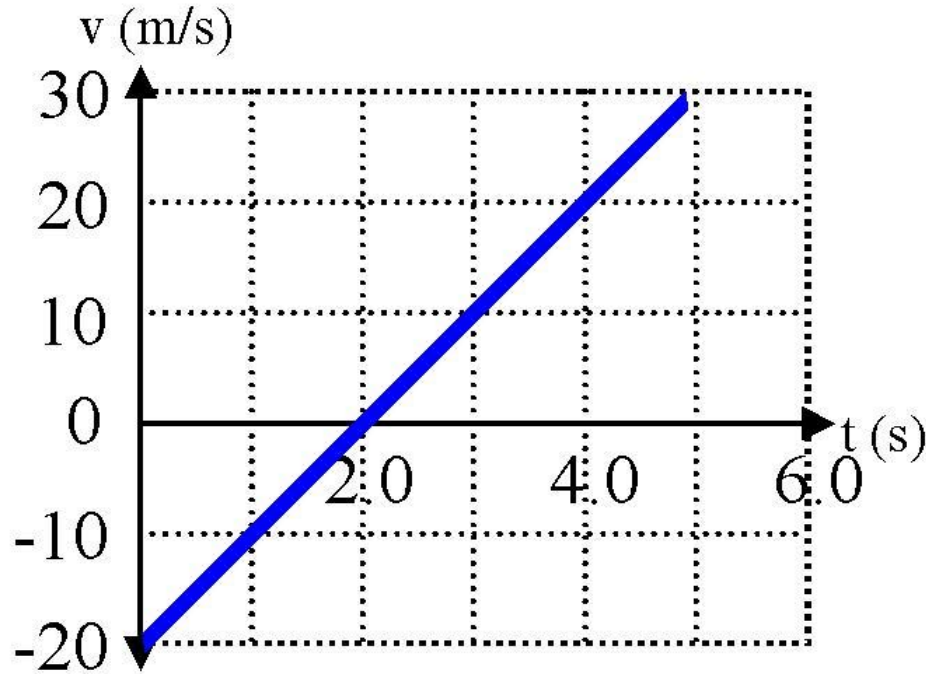
Acceleration
vs
time

AREA UNDER CURVES

The area under a velocity-time represents displacement. Distance can be calculated also; just disregard the negatives.

The area under an acceleration time graph represents the change in velocity of an object.

Practice Problem: Determine the displacement of the object from 0 to 5 seconds.



How would you describe the motion of this particle?

QUESTIONS

If acceleration is zero, what does this mean about the motion of an object?

Is it possible for a racecar circling a track to have zero acceleration?

UNIFORM (CONSTANT) ACCELERATION

In AP Physics 1, we will generally assume that acceleration is *constant*.
With this assumption we are free to use this equation:

The SI unit of acceleration is m/s².

$$a = \frac{\Delta v}{\Delta t}$$

ACCELERATION IN 1-D MOTION HAS A SIGN!

Think-Pair-Share:

- If the sign of the velocity and the sign of the acceleration is the same, what is happening to the speed of the object?
- If the sign of the velocity and the sign of the acceleration are different, what is happening to the speed of the object?

KINEMATIC EQUATIONS

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x = x_0 + [(v_0 + v) / 2]t$$

Practice Problem: A 747 airliner reaches its takeoff speed of 180 mph in 30 seconds. What is its average acceleration?

2.682 m/s²

Practice Problem: A horse is running with an initial velocity of 11 m/s, and begins to accelerate at -1.81 m/s^2 . How long does it take the horse to stop?

6.077 s

Practice Problem: A plane is flying in a northwest direction when it lands, touching the end of the runway with a speed of 130 m/s. If the runway is 1.0 km long, what must the acceleration of the plane be if it is to stop while leaving $\frac{1}{4}$ of the runway remaining as a safety margin?

- 11.267 m/s²

Practice Problem: On a ride called the Detonator at *Worlds of Fun* in Kansas City, passengers accelerate straight downward from 0 to 20 m/s in 1.0 second.

) What is the average acceleration of the passengers on this ride?

-20 m/s²

) How fast would they be going if they accelerated for an additional second at this rate?

40 m/s

Practice Problem: You are driving through town at 12.0 m/s when suddenly a ball rolls out in front of you. You apply the brakes and decelerate at 3.5 m/s².

How far do you travel before stopping?

20.571 m

PRACTICE PROBLEM

An object starts from rest and travels 1000 m reaching a velocity of 50 m/s.

What is the acceleration of the object?

1.25 m/s²

FREE FALL

Free fall is a term we use to indicate that an object is falling under the influence of gravity, **with gravity being the only force on the object.**

Gravity accelerates the object toward the earth the entire time it rises, and the entire time it falls.

The acceleration due to gravity near the surface of the earth has a magnitude of 9.8 m/s^2 . The direction of this acceleration is **DOWN**.

Air resistance is ignored.

VECTORS

Add acceleration and velocity vectors to the picture of a ball that is thrown up and falls straight back down.

On the way up



On the way down



SYMMETRY IN FREE FALL

When something is thrown straight upward under the influence of gravity, and then returns to the thrower, this is very symmetric.

The object spends half its time traveling up; half traveling down.

Velocity when it returns to the ground is the opposite of the velocity it was thrown upward with.

Acceleration is 9.8 m/s^2 and directed DOWN the entire time the object is in the air!

Practice Problem: You throw a ball straight upward into the air with a velocity of 20.0 m/s , and you catch the ball some time later.

a) How long is the ball in the air?

4 s

b) How high does the ball go?

20 m

c) What is the ball's velocity when you catch it?

-20 m/s

Practice Problem: You drop a ball from rest off a 120 m high cliff.
Assuming air resistance is negligible,

a) how long is the ball in the air?

4.900 s

b) what is the ball's speed and velocity when it strikes the ground at the base of the cliff?

-48.02 m/s